國立清華大學資訊工程系 115 學年度上學期專題報告

National Tsing Hua University, Department of Computer Science, Undergraduate Project Report, Fall 2026

專題名稱	**Beautified Is Not Forged: A Face Checker That Separates Beautification from Forgery and Shows Checkable Evidence**
參加競賽或計畫	[□ 參加對外競賽　□ 參與其他計畫　□ 無參加對外競賽或任何計畫]

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摘要	Face-forgery detectors answer “real” or “fake”. A genuine photograph that has only been through a beauty app or a retouching service is neither, and once the skin is smoothed the detectors tend to call it fake. On pixel-aligned pairs of genuine faces and their commercially smoothed renders, sixteen of seventeen published detectors score the smoothed face as more forged than its own original. A controlled training experiment shows that the cause is which class saw skin smoothing during training: showing it to both classes lowers the accusation from 54 to 12 percent. We build on that recipe a face checker that answers real, forged or beautified, marks the face part its evidence came from, and shows only sentences that code composes from the model’s output and checks before display. Restoring the marked region to its original pixels removes 94 percent of the model’s part-level forgery decisions, about twice what a post-hoc heat map achieves, and free-form descriptions written by vision-language models fail a blind matching test, so no model is allowed to describe the image. The checker runs in a web browser and the photograph never leaves the device.

中華民國 115 年 11 月